

Anti-aliasing

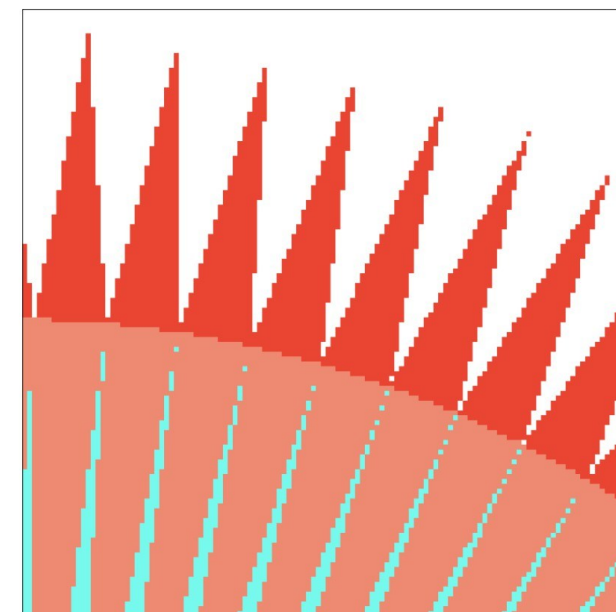
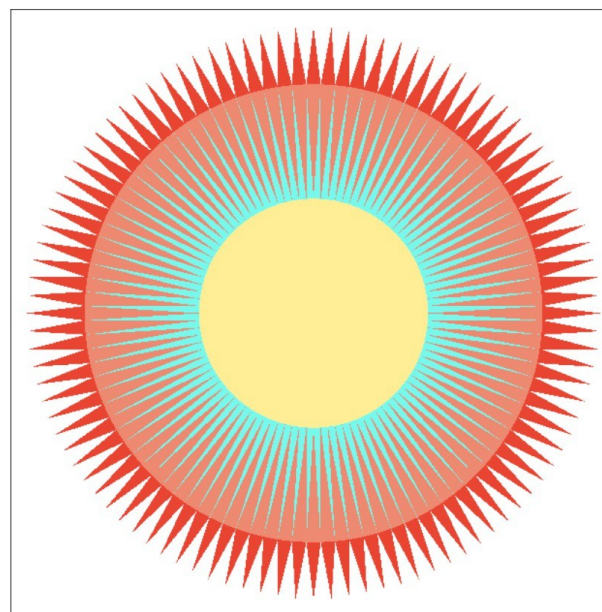
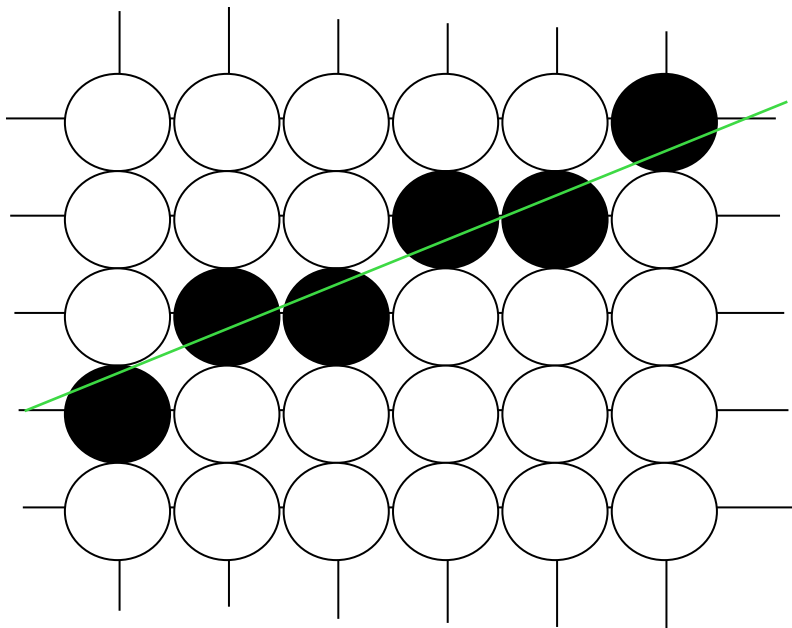
Aliasing artifacts in daily life



Aliasing (锯齿、混淆)

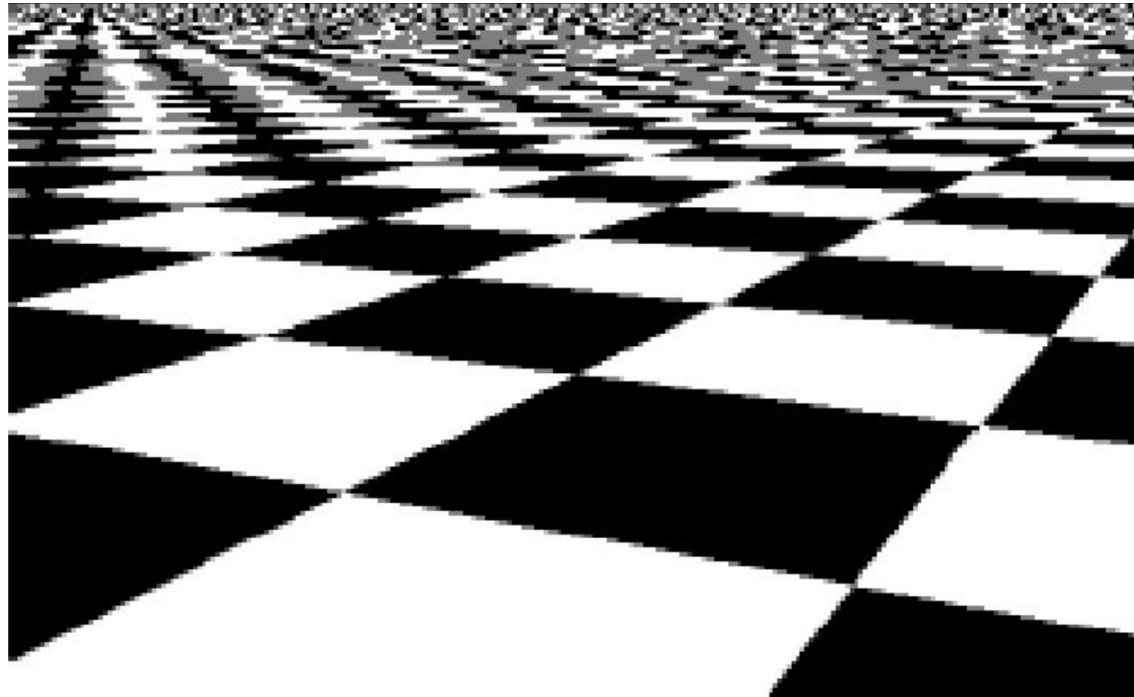
Discrete samples don't capture continuous geometry changes

- Not dense enough!



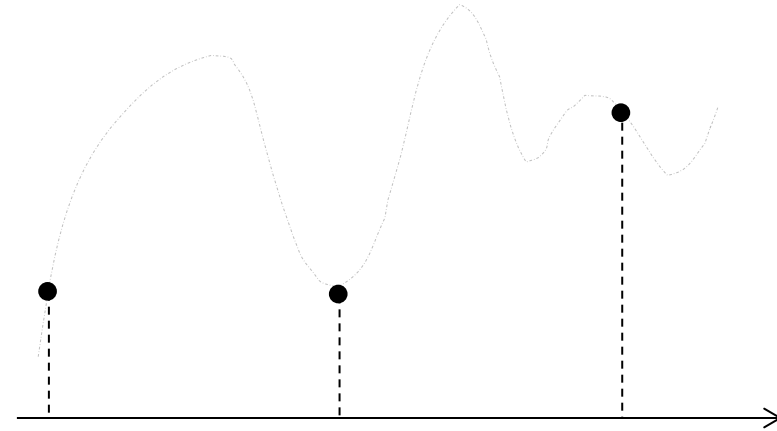
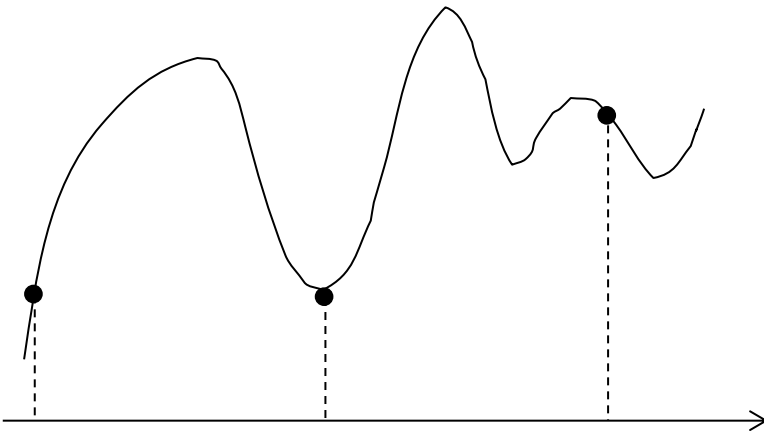
Aliasing

Can^{not} have sufficient screen resolution (sample density) for various image details, e.g., a mosaic floor near horizon

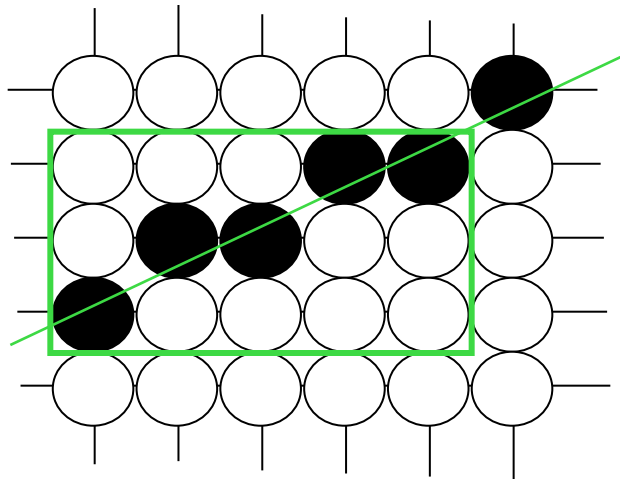


Why this happens?

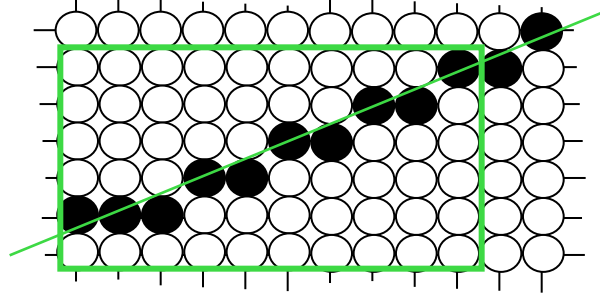
Discrete samples don't capture continuous geometry changes
- Not dense enough!



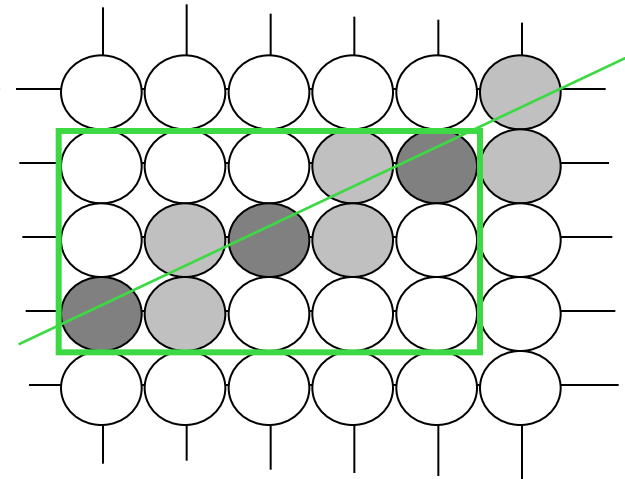
Antialiasing: super-sampling



screen
resolution



Increasing
resolution



Back to screen
resolution

Antialiasing: full-fledged tool – frequency analysis

- A signal can be decomposed into signals of various frequencies
- Transform *time domain* to *frequency domain*
- **Fourier Transform**

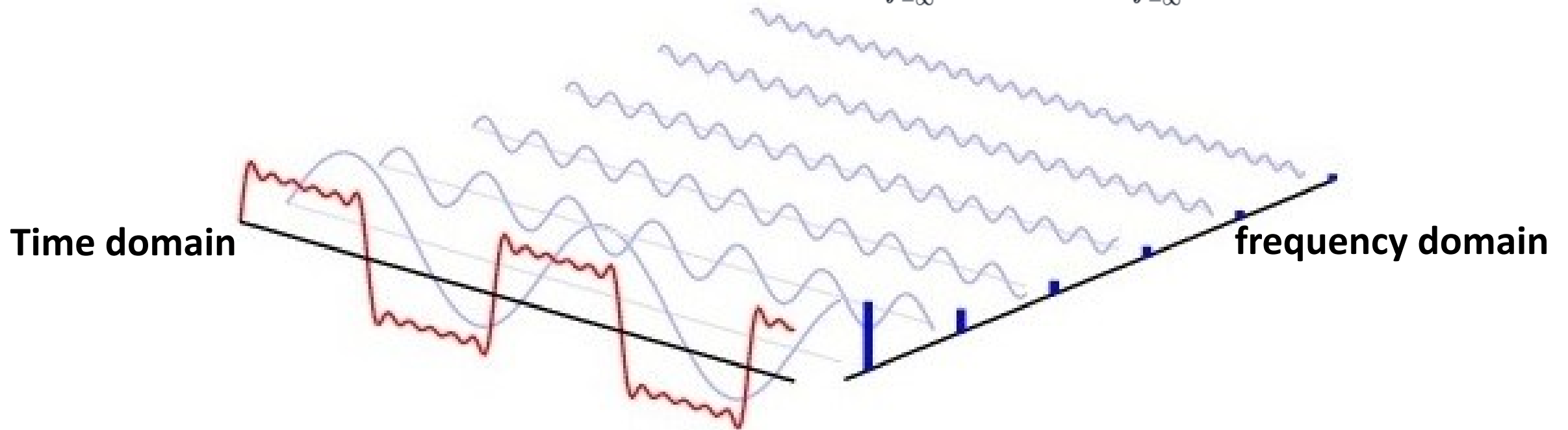
$$G(f) = \int_{-\infty}^{\infty} g(t)e^{-2\pi itf} dt, \quad \forall f \in \mathbb{R},$$

- Decompose a target signal into a combination of multiple single-frequency sinusoidal signals
- Magnitude of the sinusoid reflects the magnitude of the corresponding frequency component of the target signal

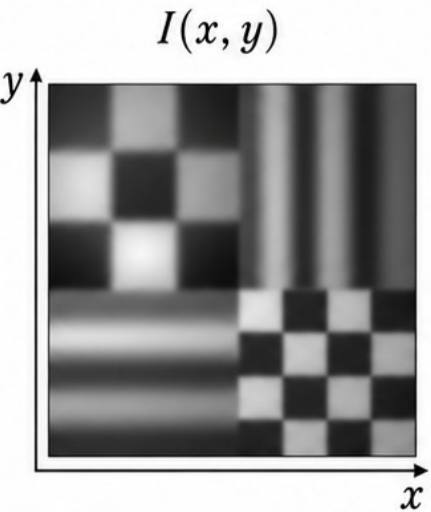
Antialiasing: frequency domain

- Transform *time domain* to *frequency domain*
- **Fourier Transform**

$$f(x) = \int_{-\infty}^{\infty} F(\omega) e^{2\pi i \omega x} d\omega = \int_{-\infty}^{\infty} F(\omega) (\cos(2\omega\pi x) + i \sin(2\omega\pi x)) d\omega$$
$$F(\omega) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i \omega x} dx = \int_{-\infty}^{\infty} f(x) (\cos(2\omega\pi x) - i \sin(2\omega\pi x)) dx$$



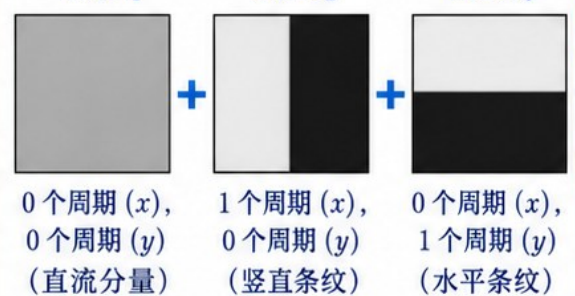
1. 原图：待分解的二维图像



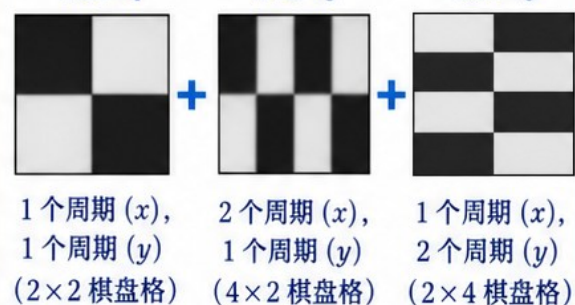
原图由若干低频和中频的棋盘格 / 条纹基底叠加而成。

2. 空间域基底（棋盘格示意）

B_1 : 频率 (0,0) 系数 a_1 B_2 : 频率 (1,0) 系数 a_2 B_3 : 频率 (0,1) 系数 a_3



B_4 : 频率 (1,1) 系数 a_4 B_5 : 频率 (2,1) 系数 a_5 B_6 : 频率 (1,2) 系数 a_6

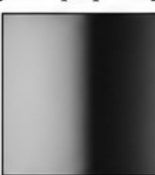


3. 累积重建

$$S_1 = a_1 B_1$$



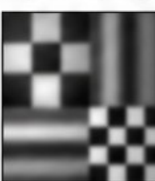
$$S_2 = a_1 B_1 + a_2 B_2$$



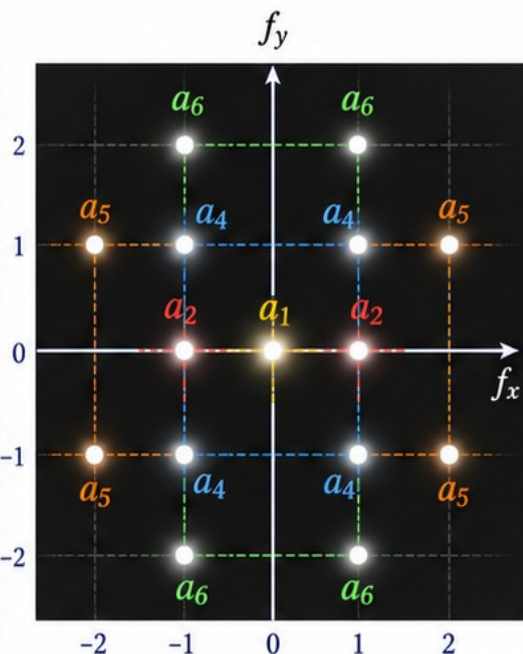
$$S_4 = \sum_{k=1}^4 a_k B_k$$



$$S_6 \approx I(x, y)$$



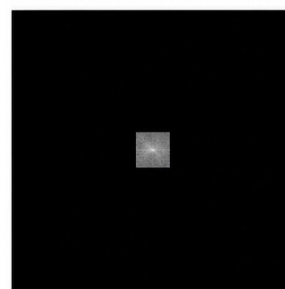
4. 对应的二维傅里叶谱 $|F(f_x, f_y)|$



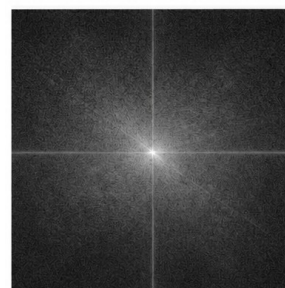
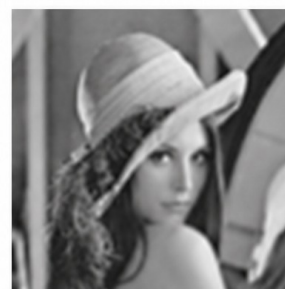
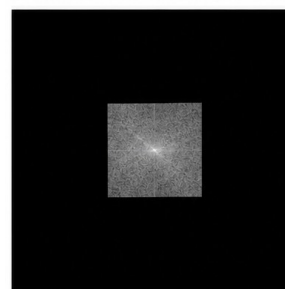
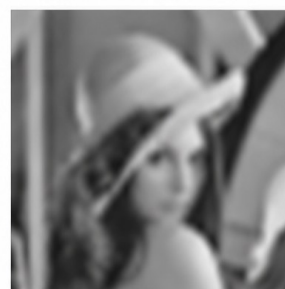
- a_1 : (0,0) — 直流分量 (中心峰)
- a_2 : ($\pm 1, 0$) — 水平频率分量 (左右对称)
- a_3 : (0, ± 1) — 垂直频率分量 (上下对称)
- a_4 : ($\pm 1, \pm 1$) — 对角频率分量 (四个对称峰)
- a_5 : ($\pm 2, \pm 1$) — (2,1) 频率分量 (四个对称峰)
- a_6 : ($\pm 1, \pm 2$) — (1,2) 频率分量 (四个对称峰)

实值图像的频谱关于原点共轭对称。

Spectrum



Image



二维傅里叶分解（空间域与频域的对对应关系）

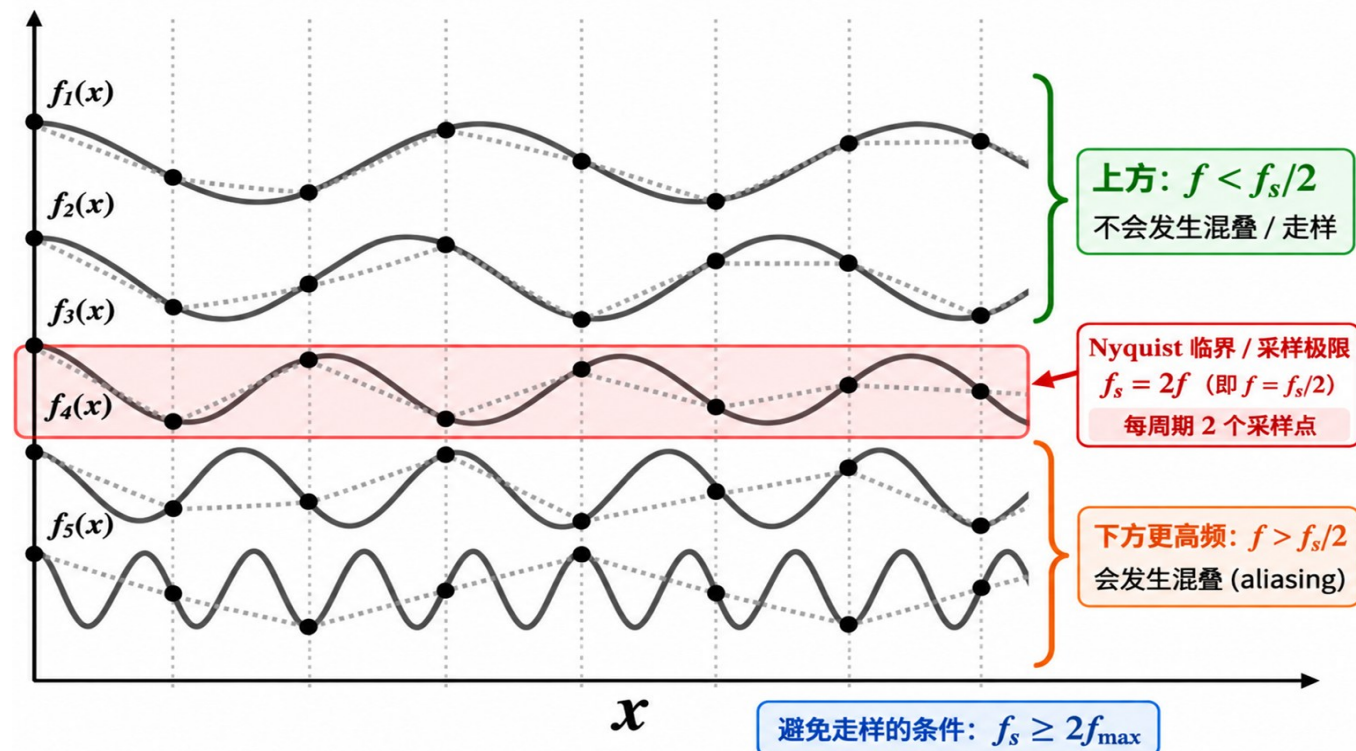
$$I(x, y) \approx \sum_{k=1}^6 a_k B_k(x, y)$$

$$F(f_x, f_y) = \mathcal{F}\{I(x, y)\}$$

对应到频域时，每个空间棋盘格基底在二维傅里叶谱上表现为若干离散峰；频率 (f_x, f_y) 决定峰的位置，系数 a_k 决定峰的强度。

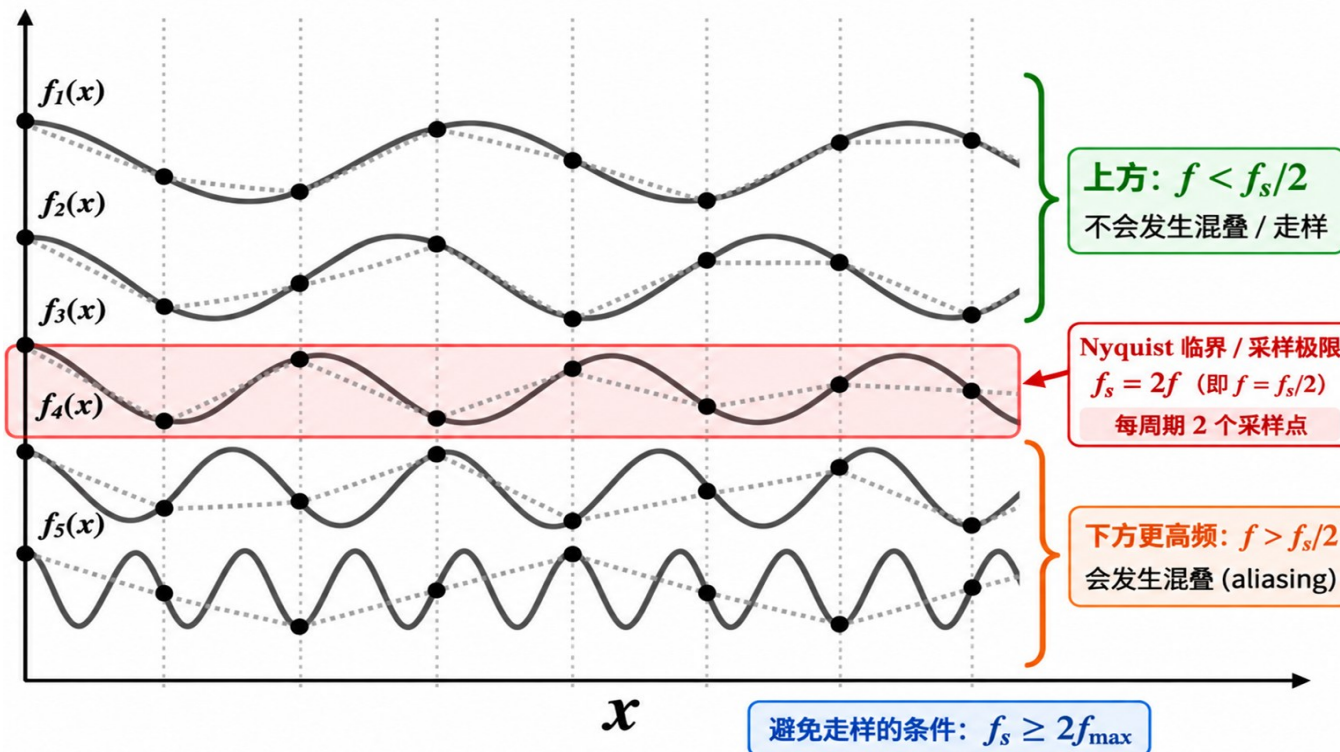
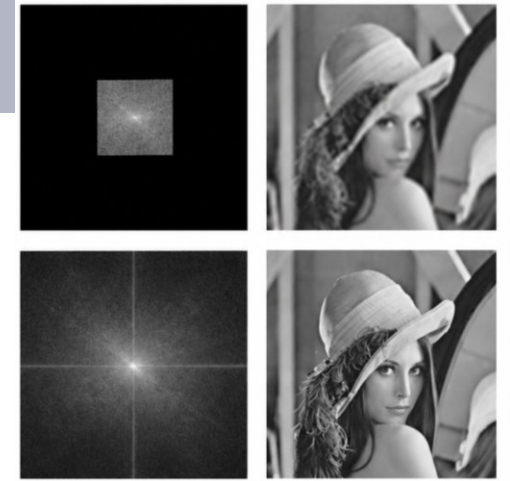
Again, why aliasing happens?

- Consider simple one-dimensional signals
- Under the same sampling frequency
 - Low frequency signals are well captured
 - Features of high frequency signals are lost



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- Consider simple one-dimensional signals
- Under the same sampling frequency
 - Low frequency signals are well captured
 - Features of high frequency signals are lost

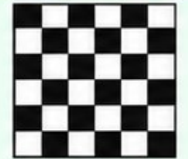


A. 单方向的 1 像素条纹

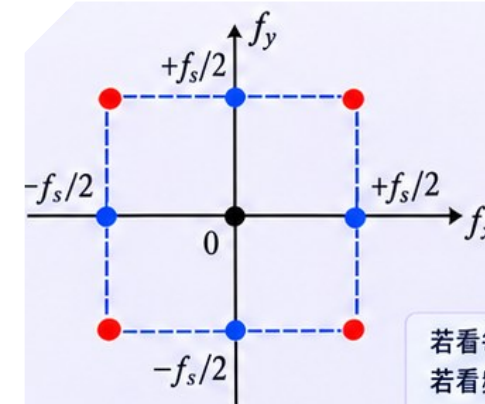


垂直条纹 (x 方向每像素交替)
频率分量: $(f_x, f_y) = (f_s/2, 0)$

B. 全像素棋盘格



黑白棋盘格 (x、y 方向都每像素交替)
频率分量: $(f_x, f_y) = (f_s/2, f_s/2)$

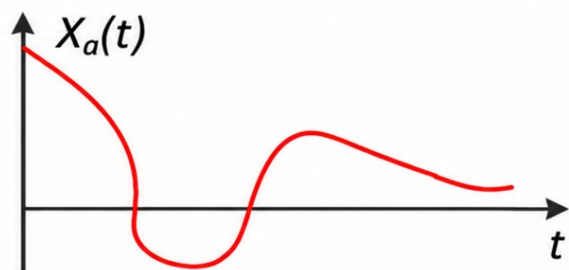


- 棋盘格峰: 在四个角上
($\pm f_s/2, \pm f_s/2$)
(x、y 两方向都达到最高频率)
- 条纹峰: 在边中点
($\pm f_s/2, 0$) 或 $(0, \pm f_s/2)$
(只在一个方向达到最高频率)

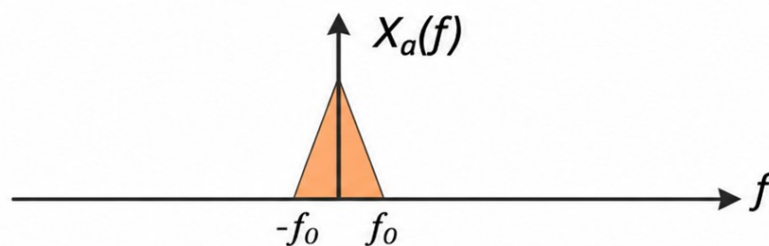
若看每个轴, 最高频率都是 $f_s/2$;
若看频谱半径, 角点的半径更大:
角点半径 = $\sqrt{(f_s/2)^2 + (f_s/2)^2} = f_s/\sqrt{2}$

Frequency domain: Periodic Repetition

- Sampling a signal in the time domain at frequency f_s is equivalent to periodically extending its frequency spectrum in the frequency domain at a period of f_s



(a)



(b)

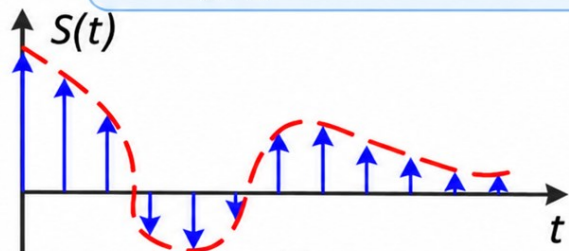
时域：采样脉冲串与原信号相乘

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s) \quad x_s(t) = x_a(t)p(t)$$

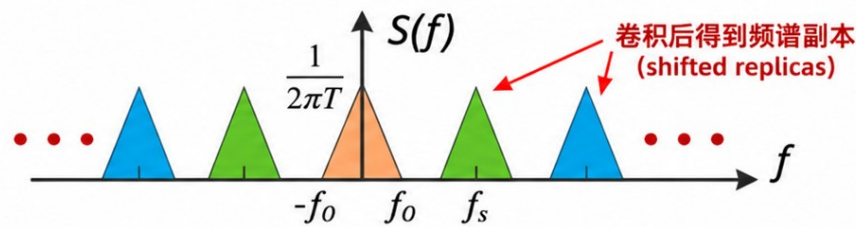
$$P(f) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} \delta(f - kf_s)$$

$$X_s(f) = X_a(f) * P(f) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X_a(f - kf_s)$$

频域：脉冲谱与原频谱卷积，产生按 f_s 间隔的平移周期延拓



(c)

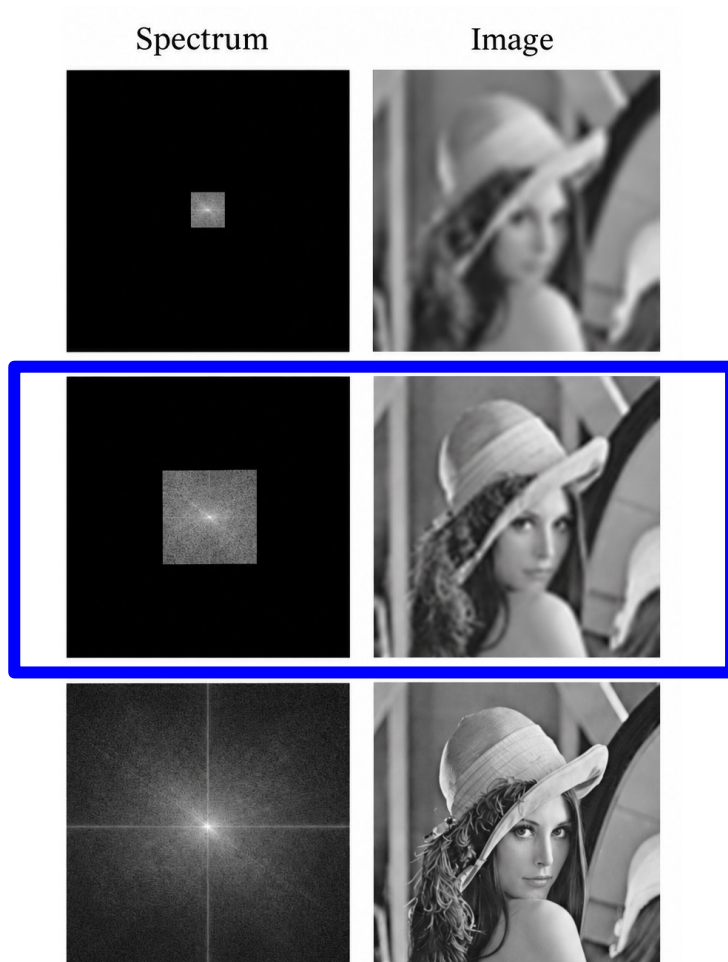
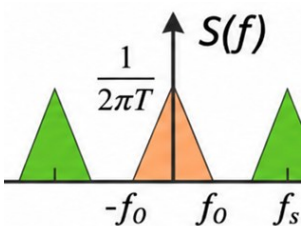


(d)

卷积后得到频谱副本
(shifted replicas)

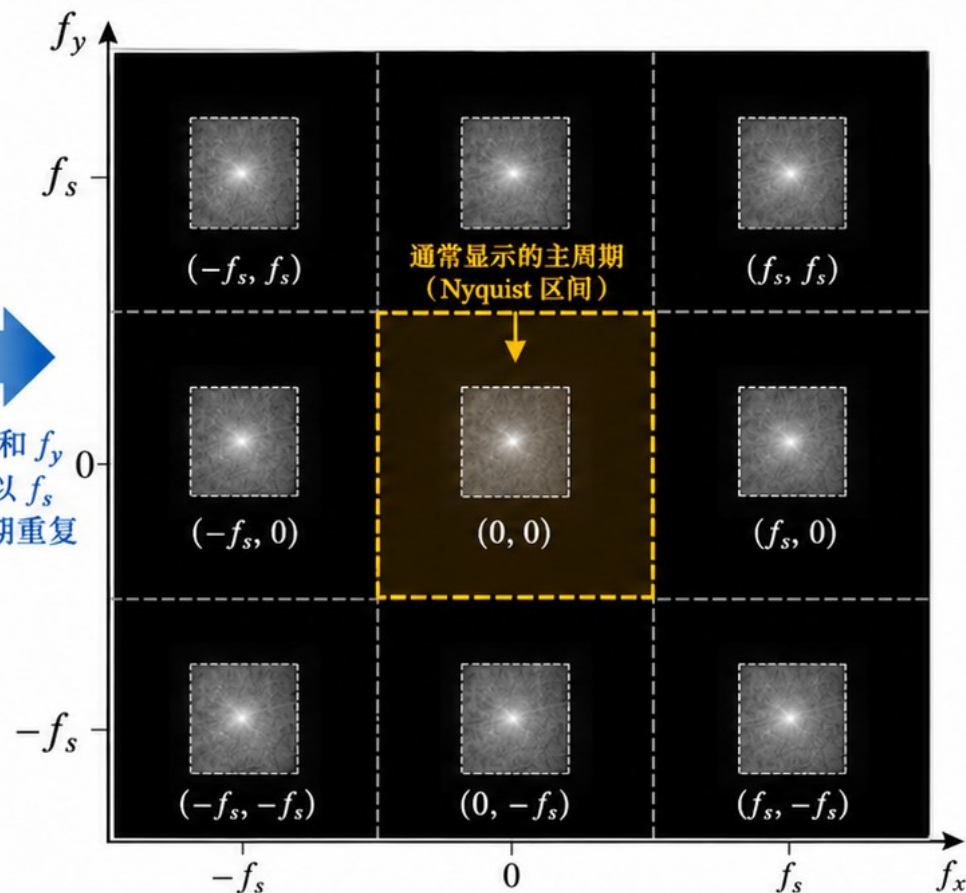
采样 → 频谱周期复制；若副本重叠则发生混叠。

Frequency domain: Periodic Repetition



沿 f_x 和 f_y
方向以 f_s
为周期重复

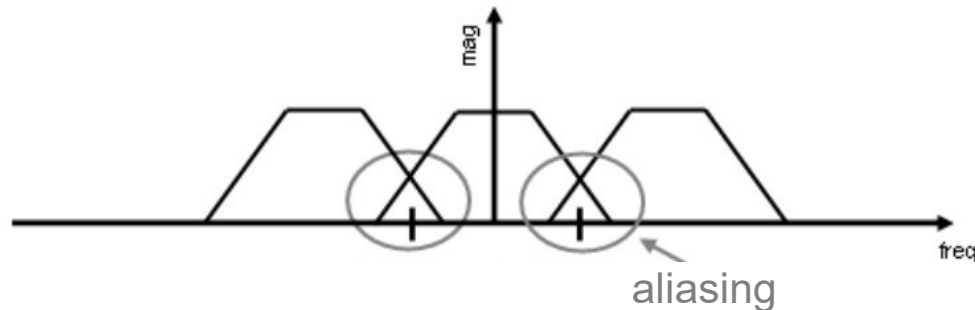
频谱周期重复 / periodic repetition



对离散图像做傅里叶变换时，我们通常只画一个基本周期（Nyquist 区间）内的频谱；如果把频谱在频率平面继续扩展，频谱会沿 f_x 和 f_y 方向以 f_s 为周期重复。

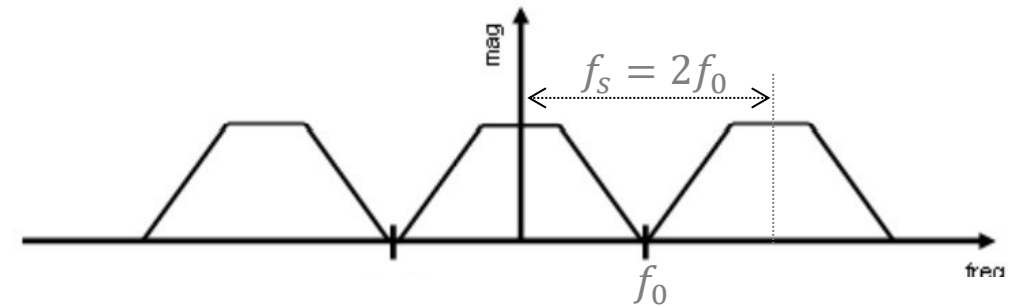
Aliasing: frequency overlay

- If the sampling frequency is too low, the period of the spectral extension will be too small, leading to the aliasing of the spectrum, also aliasing of the signal
- **Nyquist-Shannon sampling theorem**: In order to recover the signal without distortion, the sampling frequency should be greater than two times the highest of the signal, i.e., $f_s \geq 2f_0$,



A low sampling frequency leads to the aliasing.

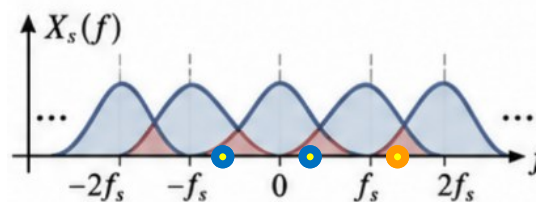
$$f_s < 2f_0$$



A high sampling frequency avoids the aliasing.

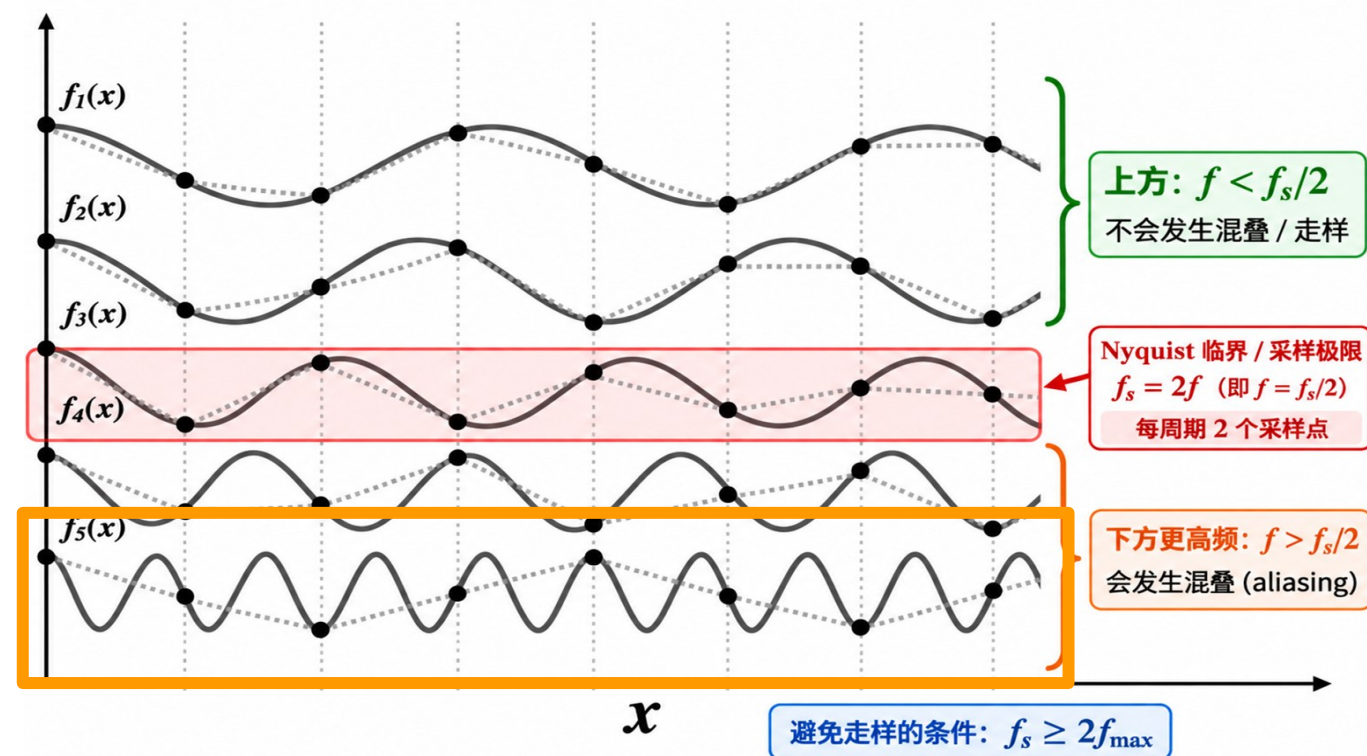
Aliasing: frequency overlay

例：若真实频率 $f = 1.25 f_s$

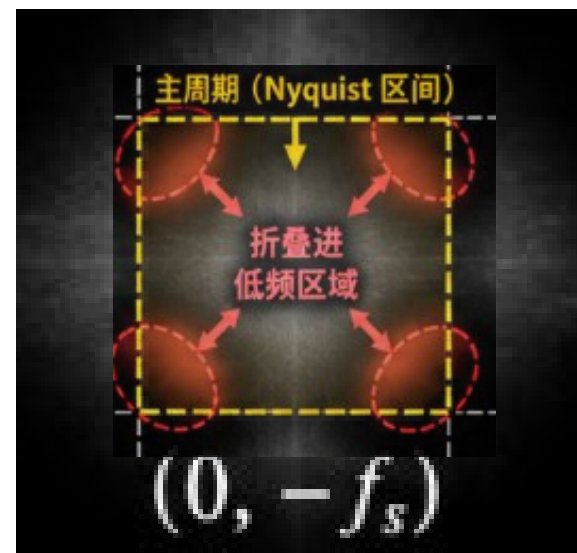
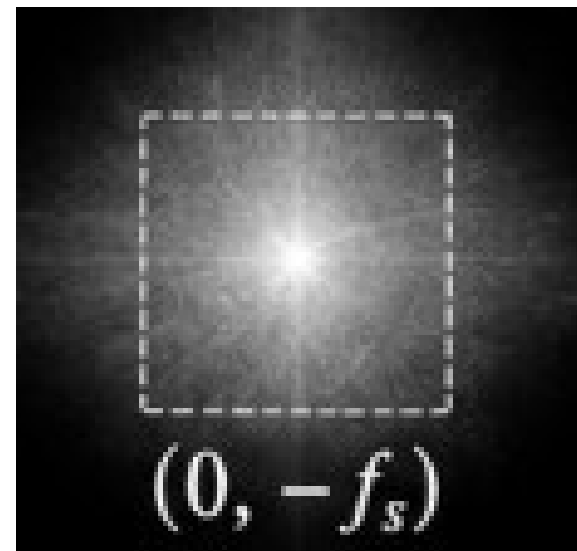


采样后看到的等效低频
(别名频率 $f_{alias} = 0.25 f_s$)

- 因此，单一频率 $1.25 f_s$ 的连续信号，在离散采样后会与 $0.25 f_s$ 的低频信号产生相同采样点 (相位/符号可有差异)，所以会表现成一个低频版本。
- 若只有这一个频率，确实会看到一个低频别名；若多个频率折叠到同一基带位置，就难以再区分原来的真实频率。

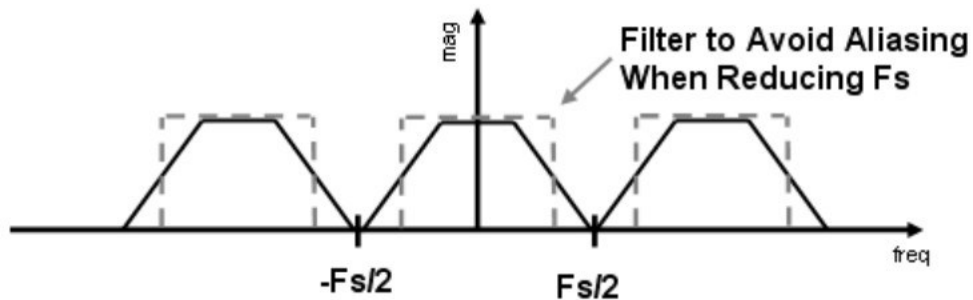


Aliasing: frequency overlay

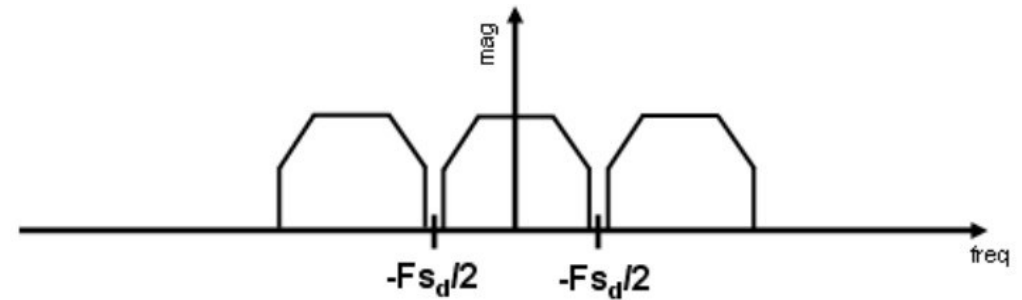


Antialiasing: low-pass filtering

- With a fixed sampling frequency (F_s), pre-process the signal to prevent aliasing
- Apply low-pass filtering to the signal, removing the component above the **Nyquist frequency ($F_s/2$)**



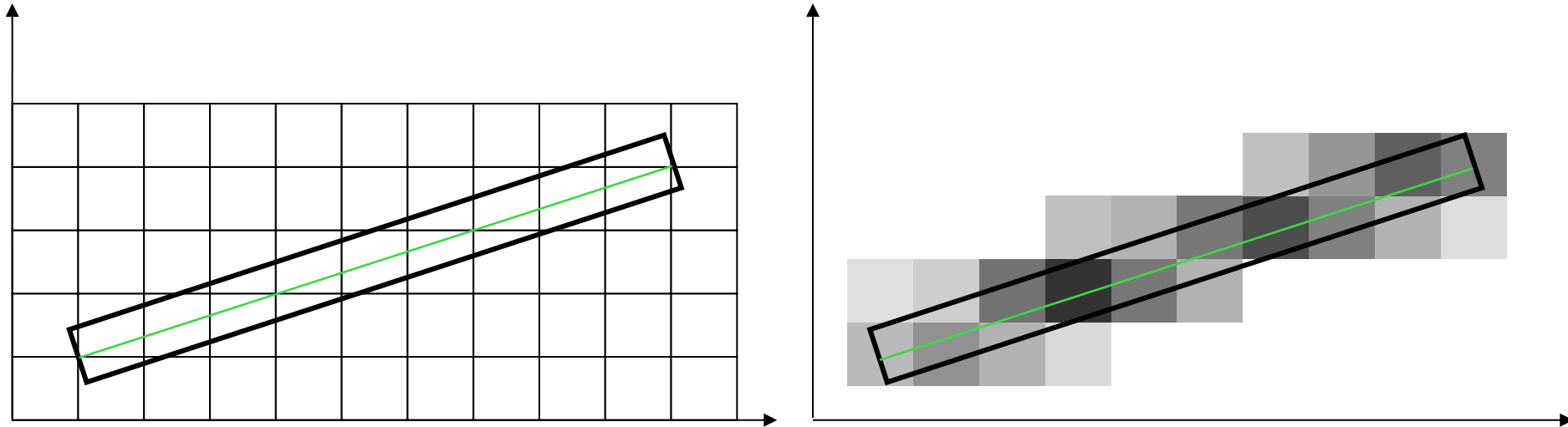
Apply low-pass filtering to the signal



Filtered signal avoids the aliasing

Antialiasing: (un)weighted area sampling

Directly computing the 'shade' value (instead of simply 0-1 sampling) is a way of low-pass filtering!



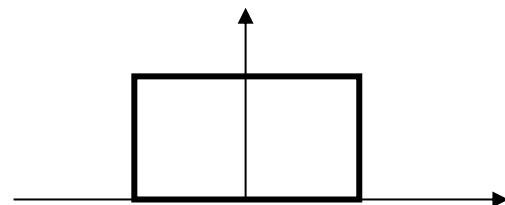
- Line with '**thickness**'
- Pixel's color, here 'blackness', depends on **the intersection area** between the thick line and the pixel square
- **Weighted area**: Intensity solely based on intersection area, equal areas equal intensity
- **Unweighted area**: 'blackness' = $\text{area} \cdot f(\text{distance})$, Where f : weighting function (aka low pass filter), dist: pixel center distance to the line.

Antialiasing: low-pass filtering function

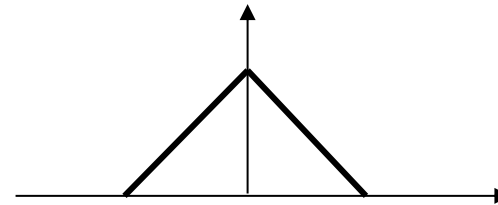
- Basically two strategies: (1) low-pass filtering **before** sampling vs. (2) low-pass filtering **after** supersampling (introduced earlier)
- The (un)weighted area sampling is one way of making a thin line wider and less sharp (0-1 appearance), which is effectively blurring (low-pass filtering effect)
- ***We can define many low-pass filtering functions!***

Can be anything, BUT,

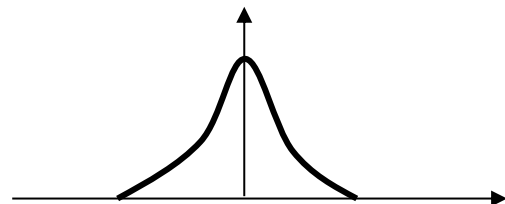
1. Finite non-zero region
2. Meaningful (e.g., decreasing from high to zero value when distance increases)
3. Full 'area' equal to 1



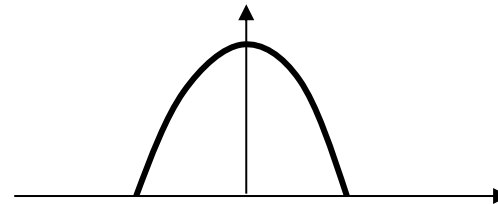
Box



Tent/Bartlett

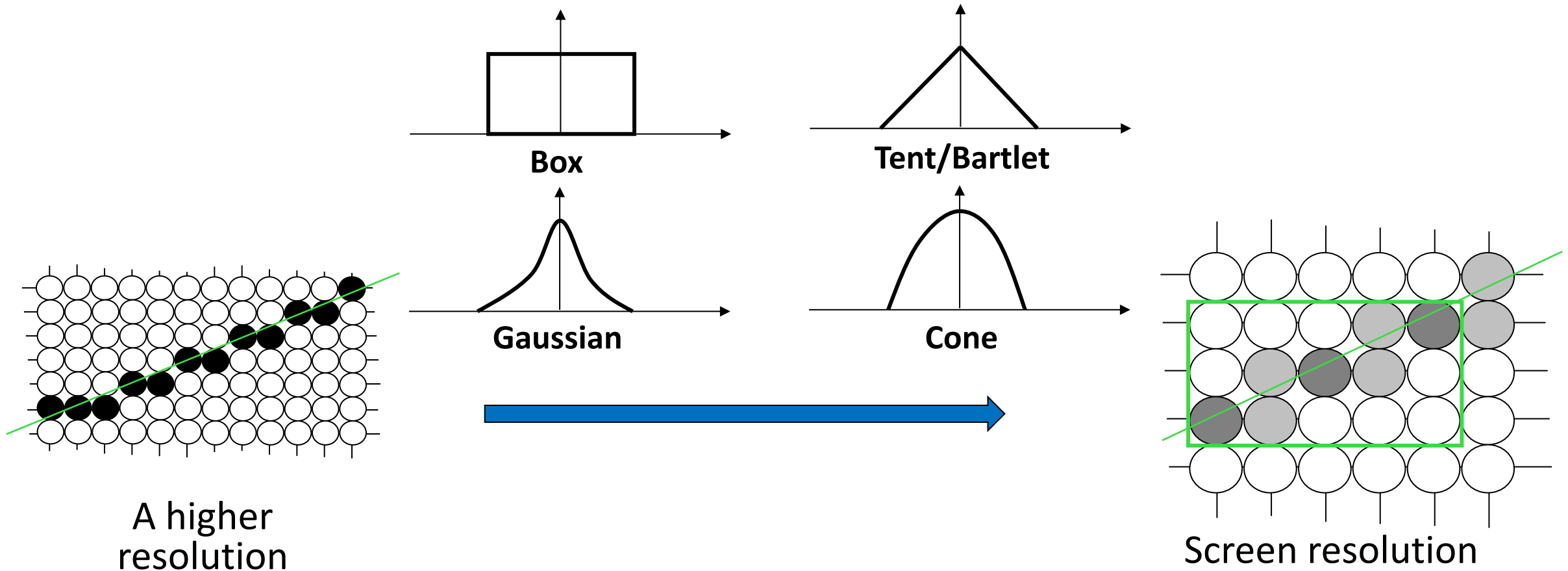


Gaussian

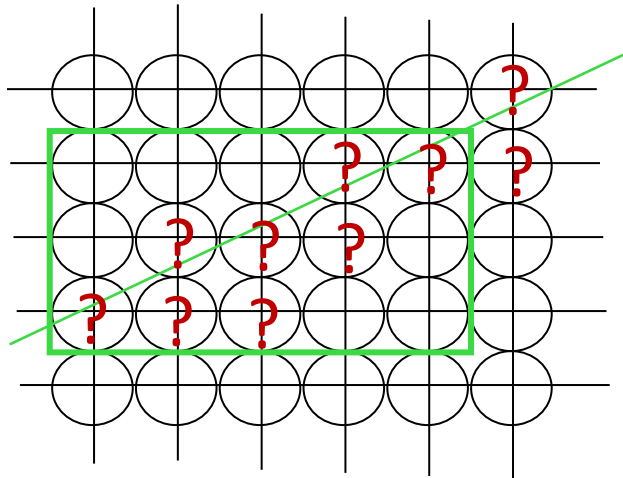


Cone

Low-pass filtering used in the earlier method



Antialiasing: low-pass *before* using distance function

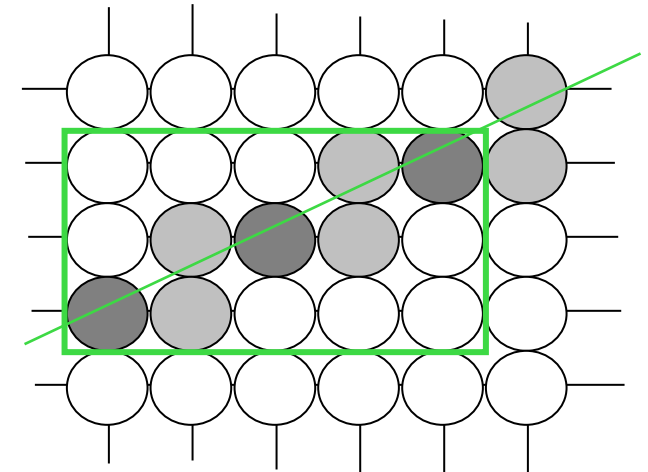


screen resolution

Compute the 'shade' value at each pixel using the aforementioned low-pass filter based on its distance to the line.



The result is very similar to that of the earlier low-pass *after* method.



screen resolution

Aliasing in image resampling

Aliasing appears not only in drawing shapes, but also in drawing (warping) images

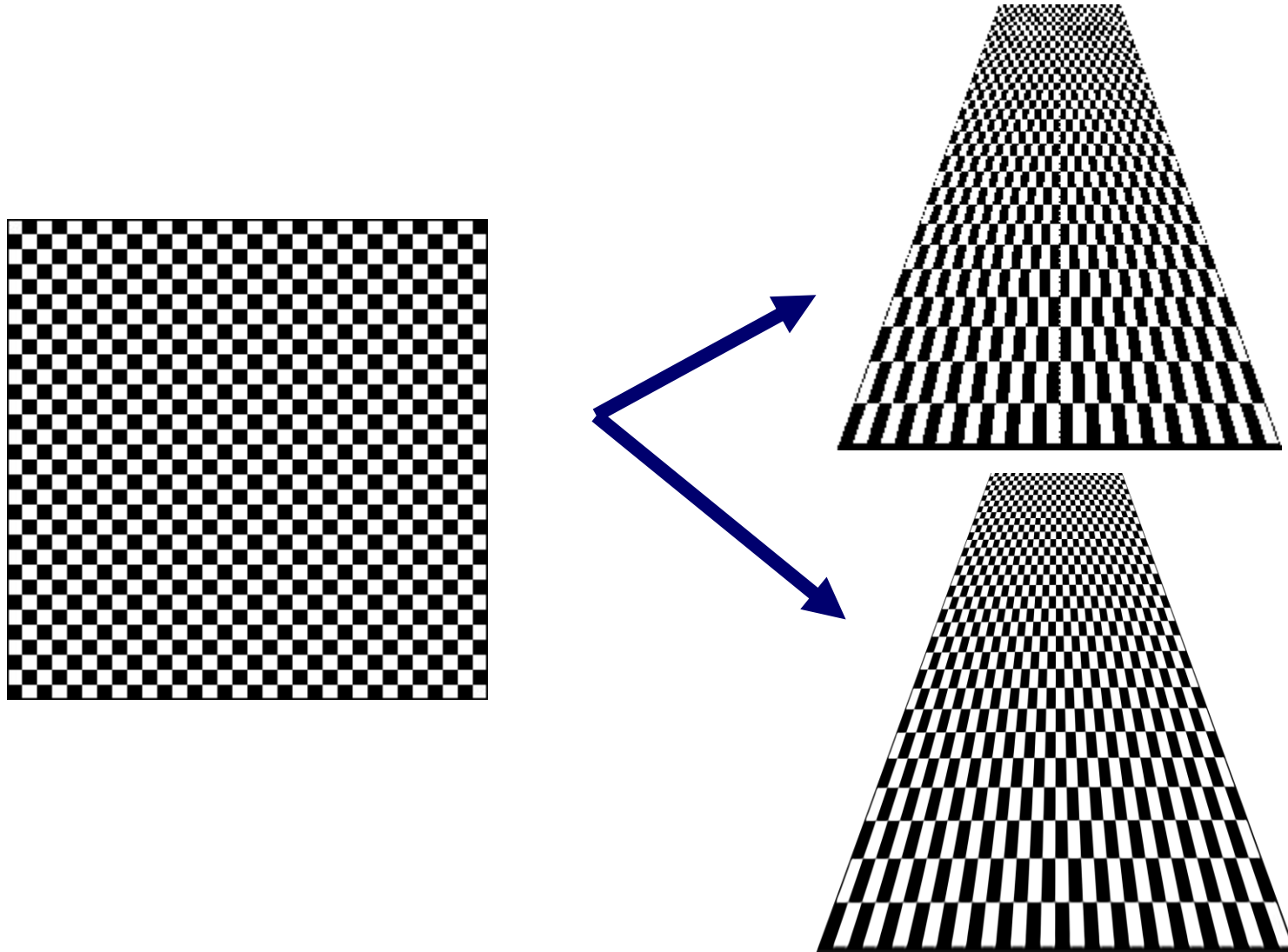
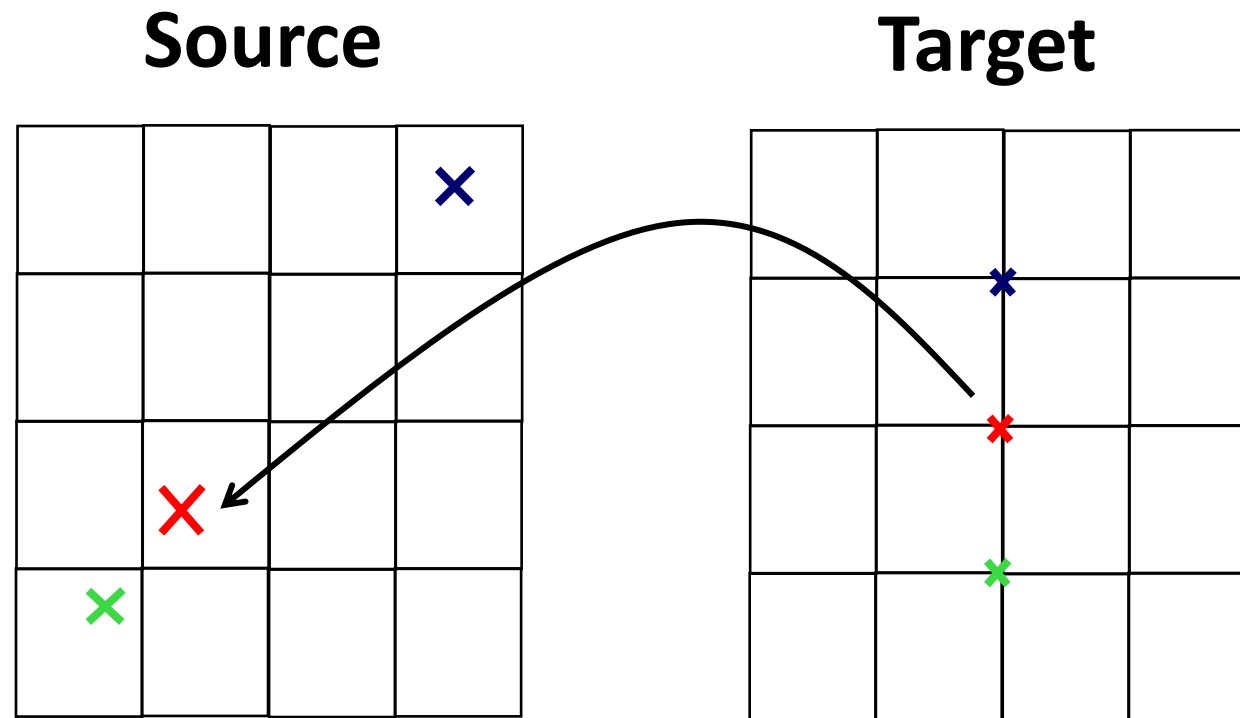


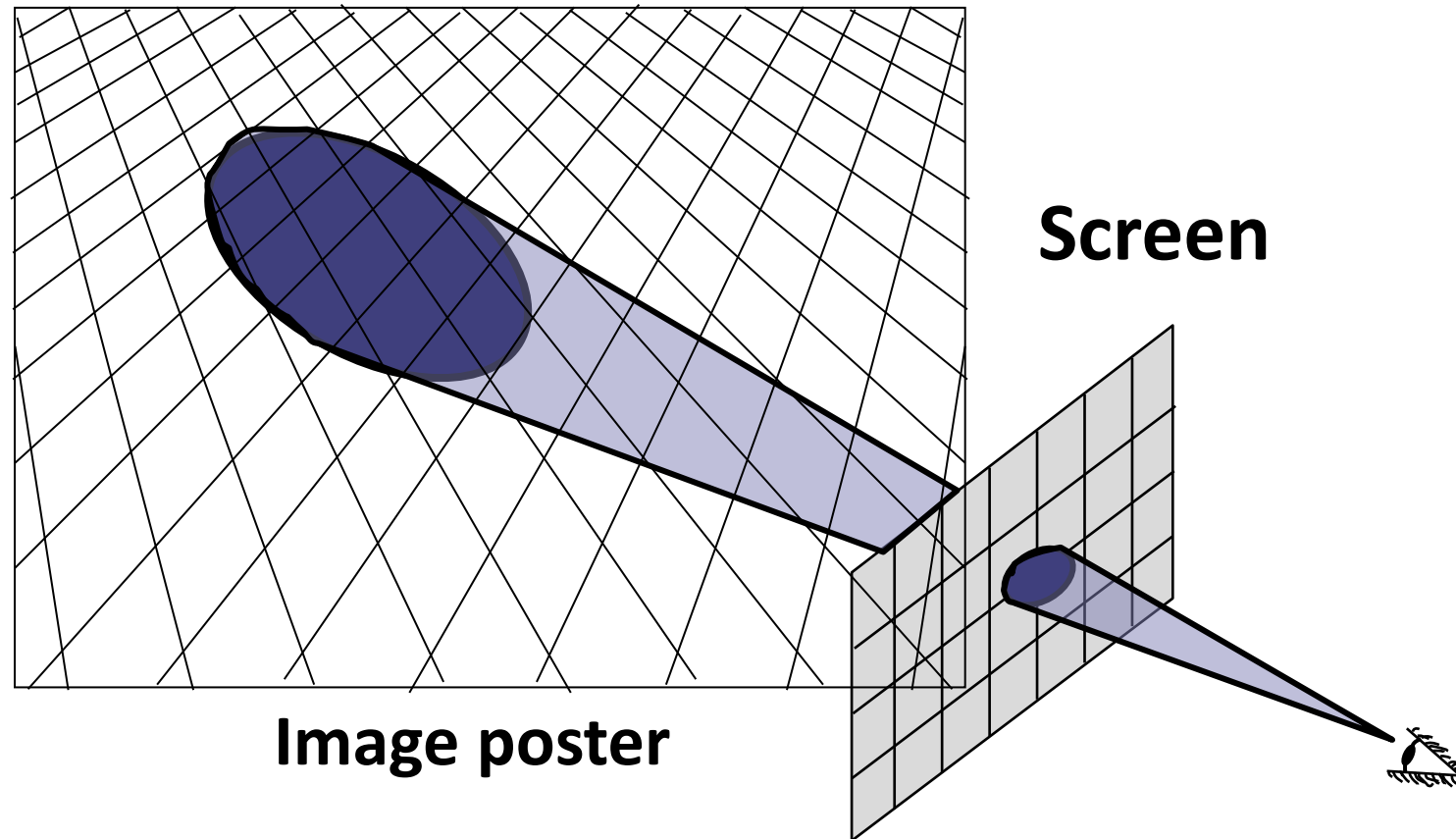
Image Mapping

- Problems arise when the warped image has different pixel resolution than that of the source image. But this varies from area to area!
- Aliasing appears when one pixel in target image corresponds to a lot of pixels in the source image (undersampling!)



Put it in perspective, literally

- Illustrates how one pixel in target image corresponds to a lot of pixels in the image poster towards to the horizon

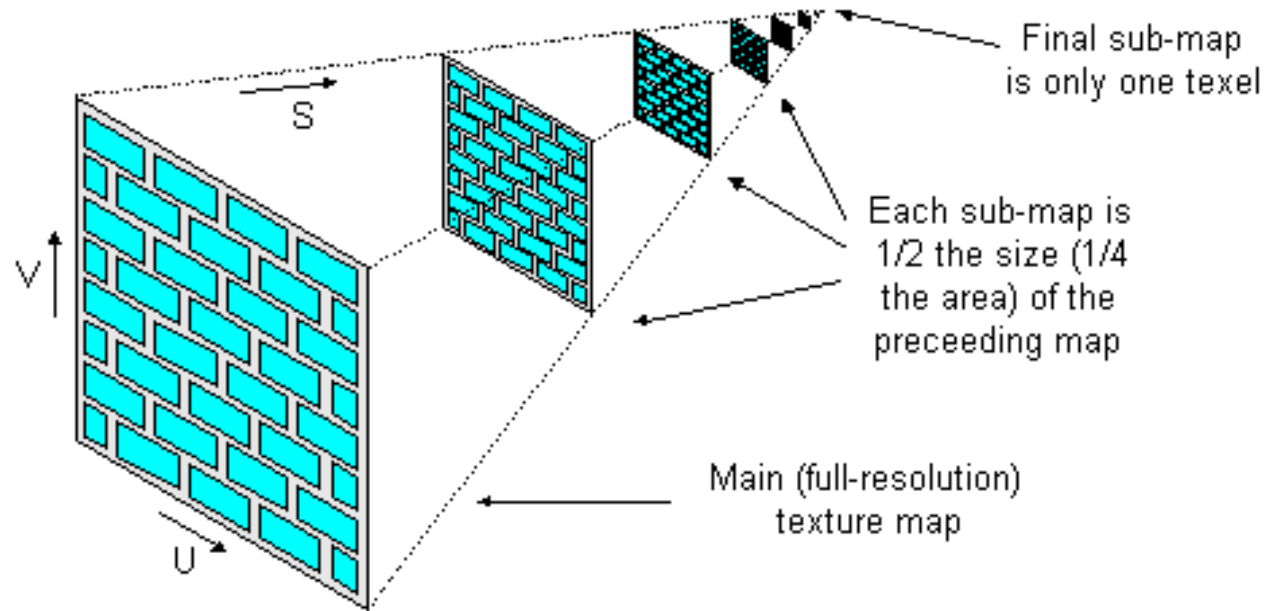


There will be more issues in 3D, later!

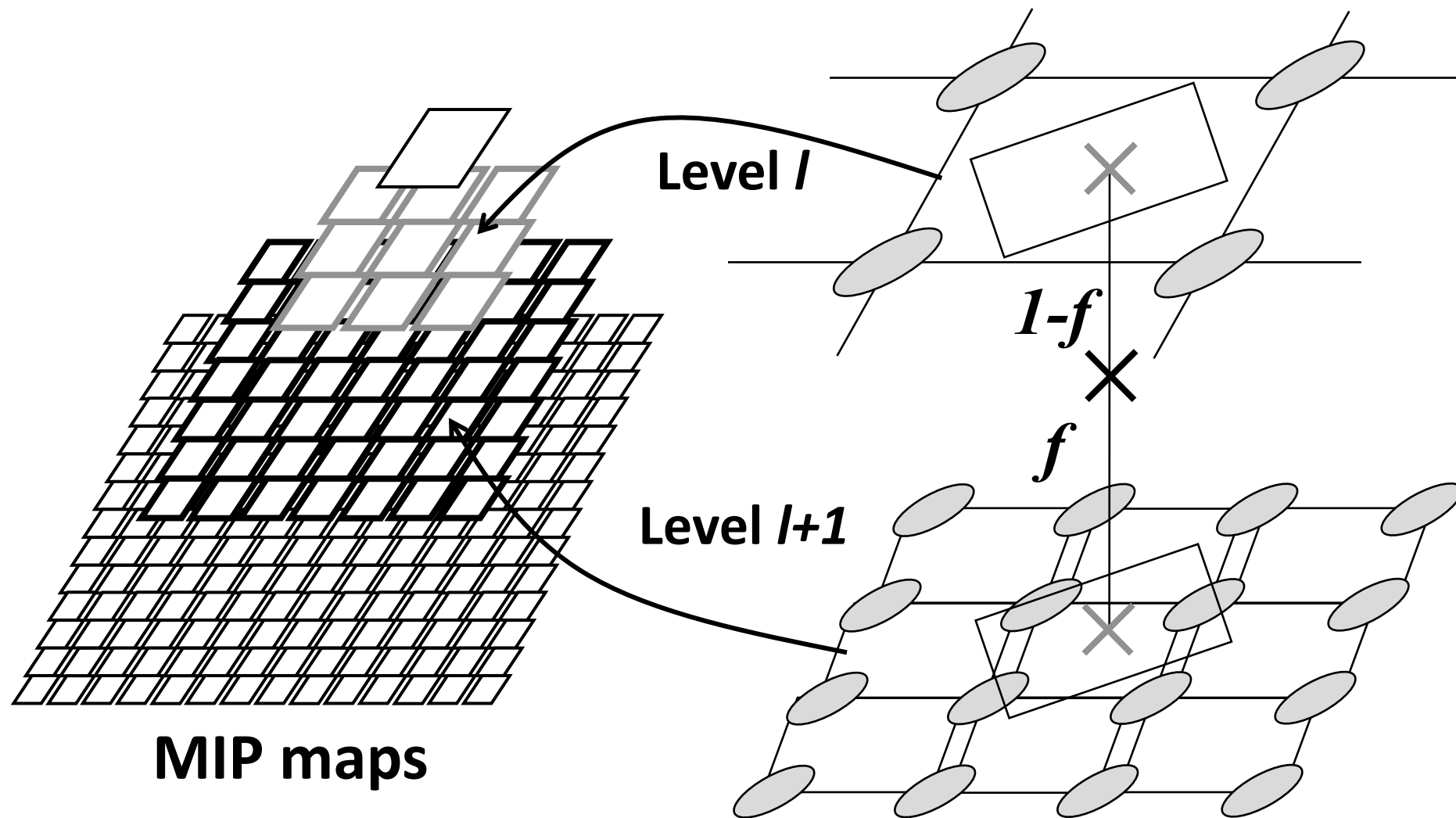
MIP Mapping

- MIP Mapping is one popular technique for antialiasing in texture mapping. MIP is an acronym for the latin phrase *multum in parvo*, which means "many in a small place". The technique was first described by Lance Williams. The basic idea is to construct a pyramid of images that are prefiltered and resampled at resolutions that are a binary fractions ($1/2$, $1/4$, $1/8$, etc) of the original image's resolution.

- While rasterizing we compute the index of the image pyramid level that has resolution *closest* to that of our desired screen resolution; in practice, two closest levels, rather than only one, are picked up and an interpolation between the two levels is performed.



MIP Mapping



Comparison



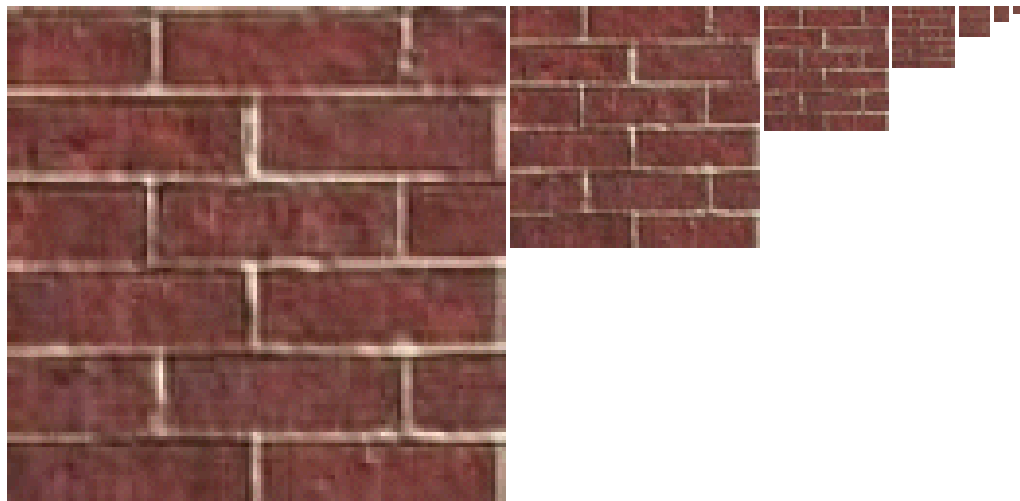
Nearest neighbor interpolation



MIP-mapping

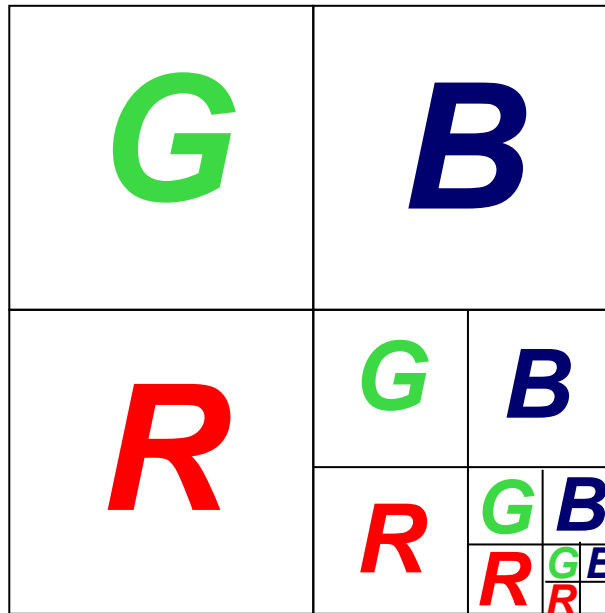
MIP Mapping

Computing this series of filtered images requires only a small fraction of additional storage over the original texture (How small of a fraction?).



Storing MIP Maps

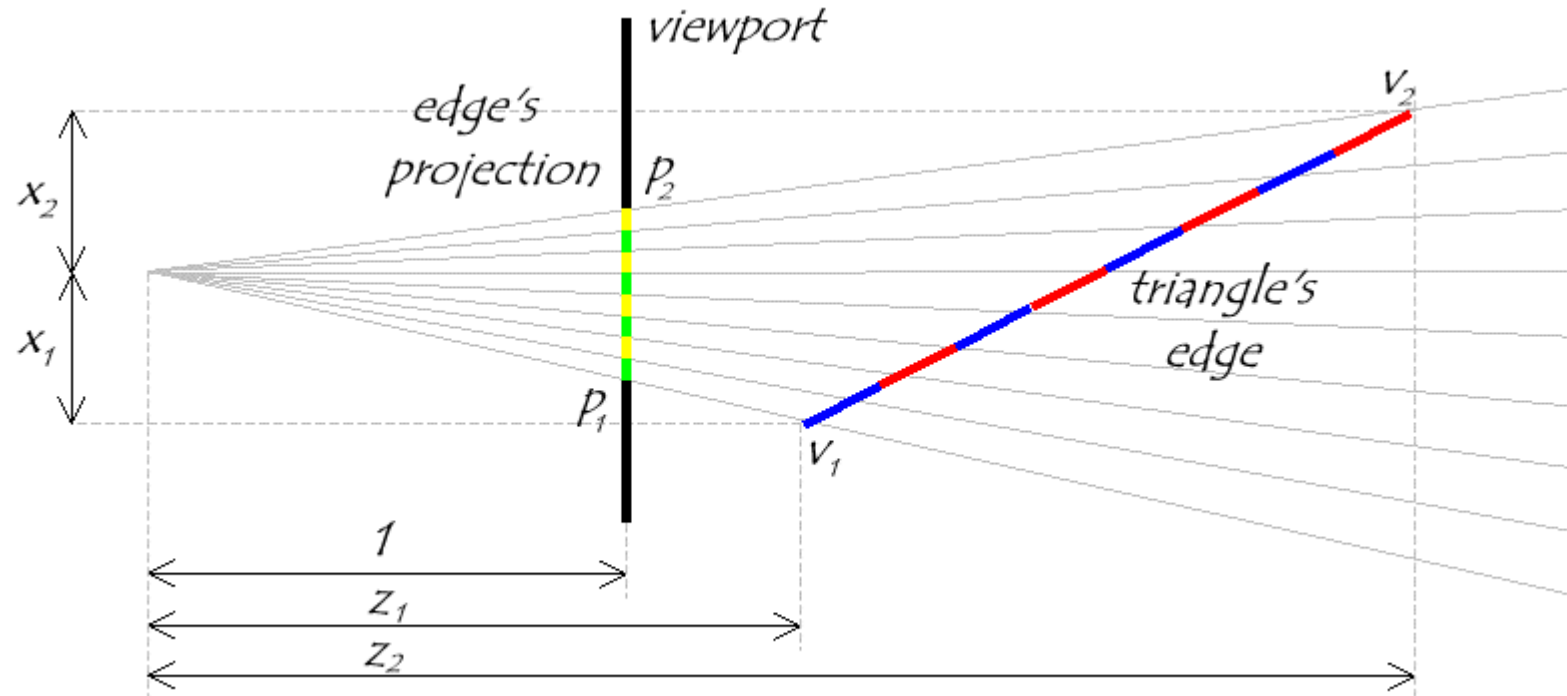
One convenient method of storing a MIP map is shown below (It also nicely illustrates the 1/3 overhead of maintaining the MIP map).



We must make a few small modifications to our rasterizer to compute the MIP map level. Remember the equations that we derived last lecture for mapping screen-space interpolants to their 3-space equivalent.

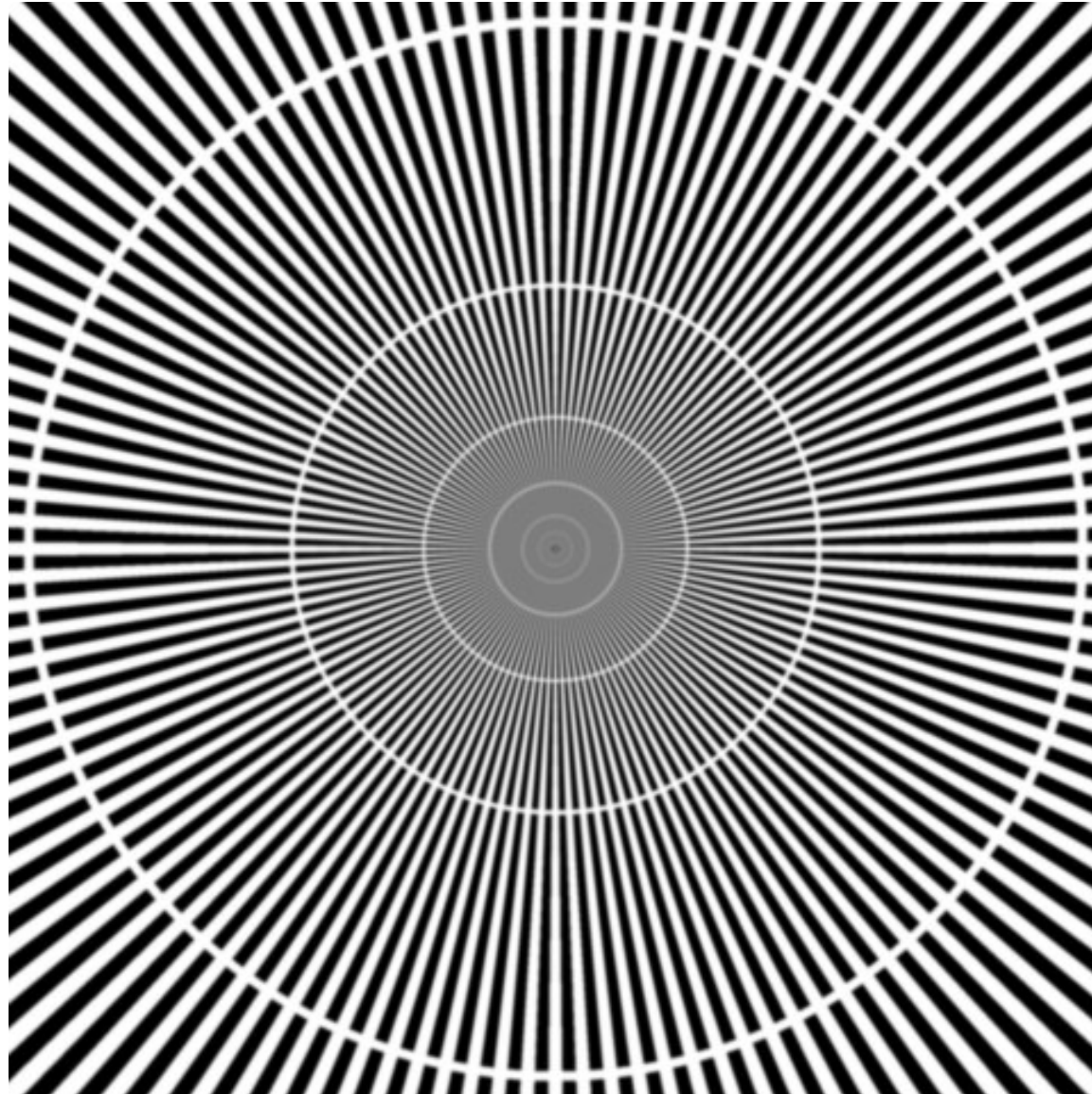
Computing MIPMAP Level - Visualizing the Problem

We will discuss this later.

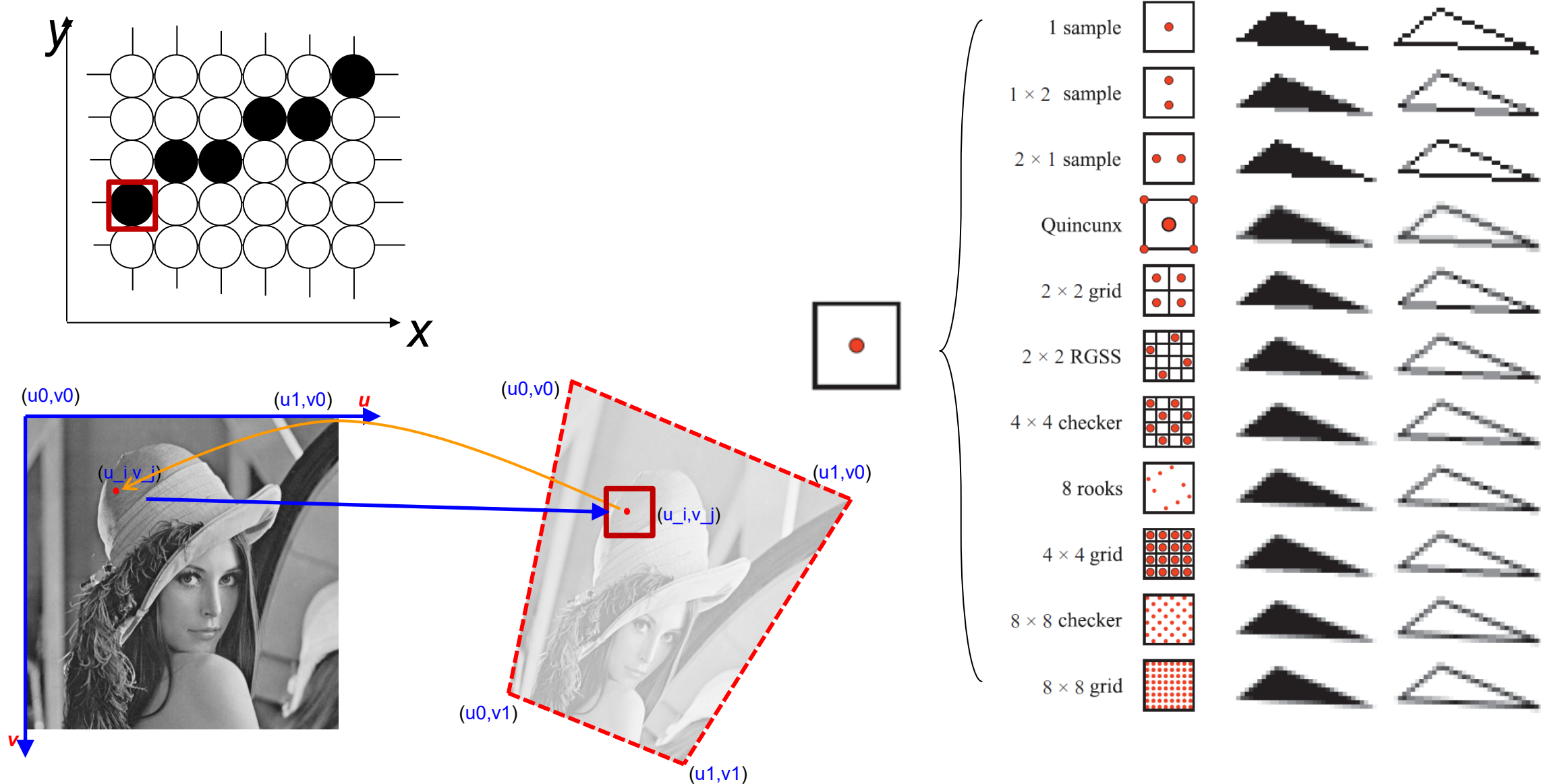


Notice that uniform steps on the image plane do not correspond to uniform steps along the edge.

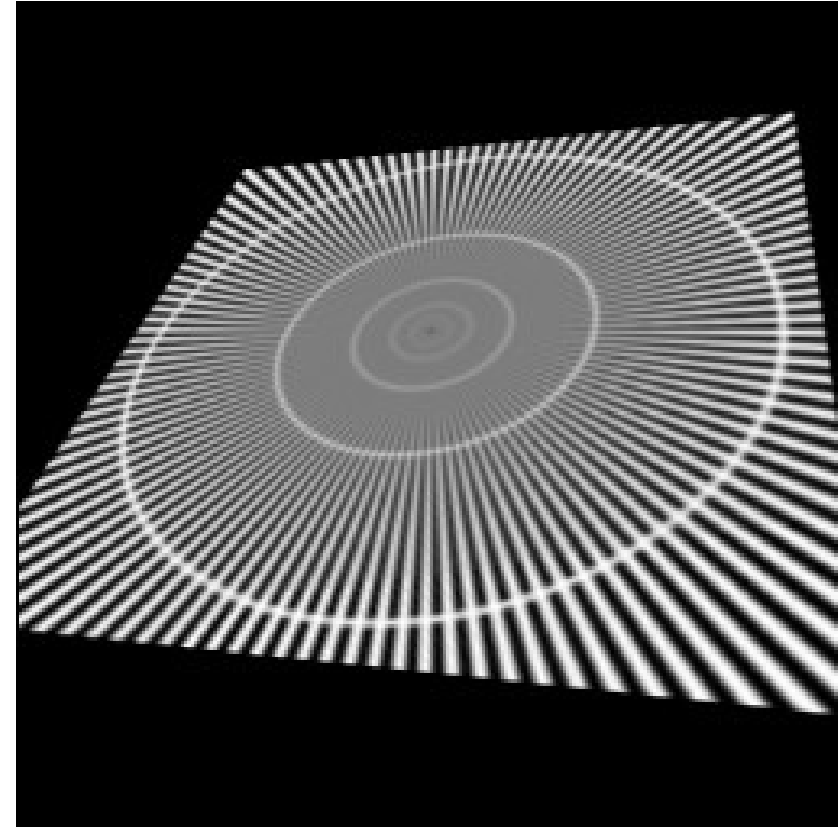
Again, How to Draw This?



Super sample anti aliasing, SSAA



Anti aliasing during Rendering



Multi Sample Anti Aliasing, MSAA

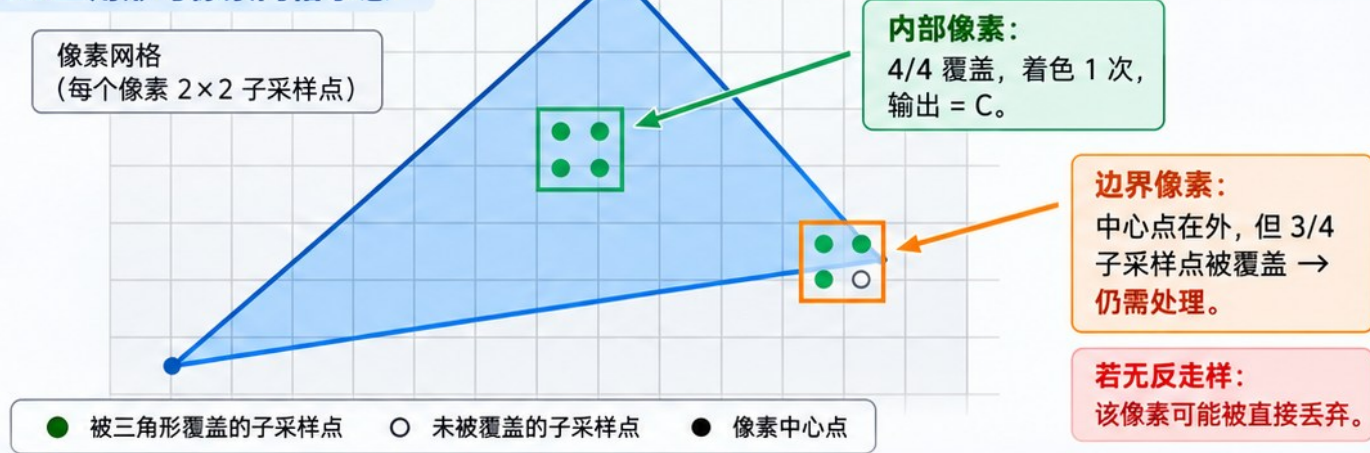
多重采样反走样 (MSAA)

内部像素仍只着色一次；边界像素通过多个子采样点估计覆盖率，从而降低锯齿。

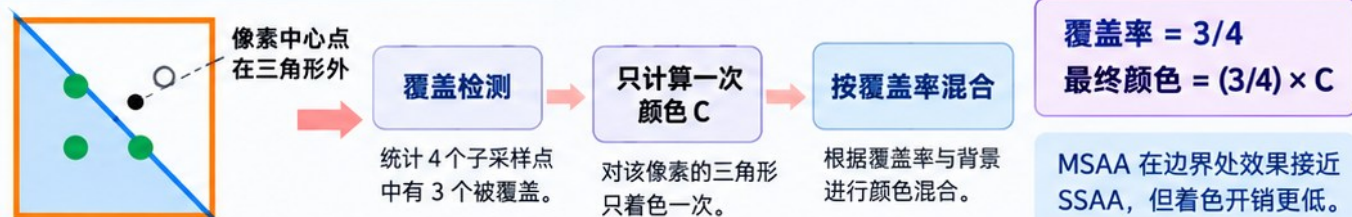
核心思想

- 1 **普通单采样**：通常只看像素中心是否落在三角形内。
- 2 **MSAA**：每个像素布置 4 个子采样点 (2×2)。
- 3 若像素中心不在三角形内，但部分子采样点被三角形覆盖，则该像素仍会被标记为需要处理。
- 4 与 SSAA 不同：MSAA 对该像素**只计算一次三角形颜色**，不对每个子采样点都单独着色。
- 5 最终像素颜色 = 覆盖率 × 颜色，例加 3/4 覆盖时，输出 = $3/4 \times C$ 。

A. 三角形与像素网格示意



B. 放大显示边界像素的 2×2 子采样点



MSAA vs SSAA

- SSAA:** 每个子采样点都单独着色，质量高，开销大。
- MSAA:** 主要对可见性做多重采样，对颜色通常只着色一次，开销更低。

直观理解

- 三角形内部:** 处理方式几乎与单采样相同。
- 三角形边界:** 根据子采样覆盖比例平滑边缘。

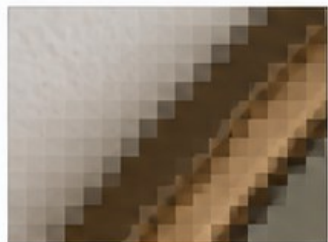
Can MSAA Improve Both?

边缘属于几何 aliasing；镜中反射属于着色 aliasing

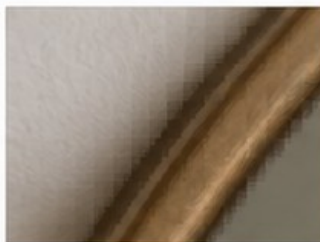
几何反走样：圆形边缘 / 轮廓边界

圆形的外轮廓属于几何 aliasing，光栅化时会出现边缘锯齿。

MSAA 擅长处理覆盖率 / 边界锯齿。



无反走样（锯齿明显）

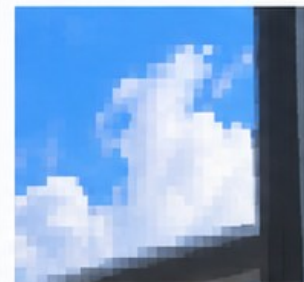


几何反走样后
（边缘平滑）

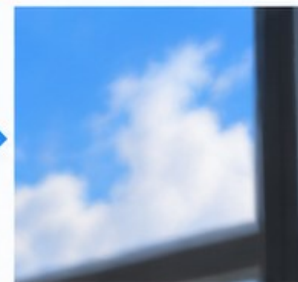
着色反走样：镜中反射 / 三角形内部颜色变化

镜中的天空、云朵、窗框等在一个三角形内部会有高频的颜色、纹理和光照变化，也会产生 aliasing，即使远离物体轮廓边缘。

高频反射、纹理、光照变化需要着色采样 / 滤波。



无反走样
（高频细节锯齿）



着色反走样后
（细节平滑）